

WASP-11b

R-0872

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_181126 · created 2026-07-28T23:37:19+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458448.6617 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.051
Transit depth (Rp/Rs)^2	3.11 +/- 1.81 [%]
Orbital Inclination (inc)	89.94 +/- 1.7
Airmass coefficient 1 (a1)	0.93 +/- 0.55
Airmass coefficient 2 (a2)	0.08 +/- 0.27
Scatter in the residuals of the lightcurve fit is	62.16 %
Best Comparison Star	#1 - [27, 160]
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.106 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.051 vs 0.131 (deviation 0.88σ)
Duration fit vs published	0.106 d vs 0.1075 d (deviation 0.15σ)
Mid-time O–C	-3.7 min
Depth SNR	3.5
Residual scatter	62.16 %

Light curve

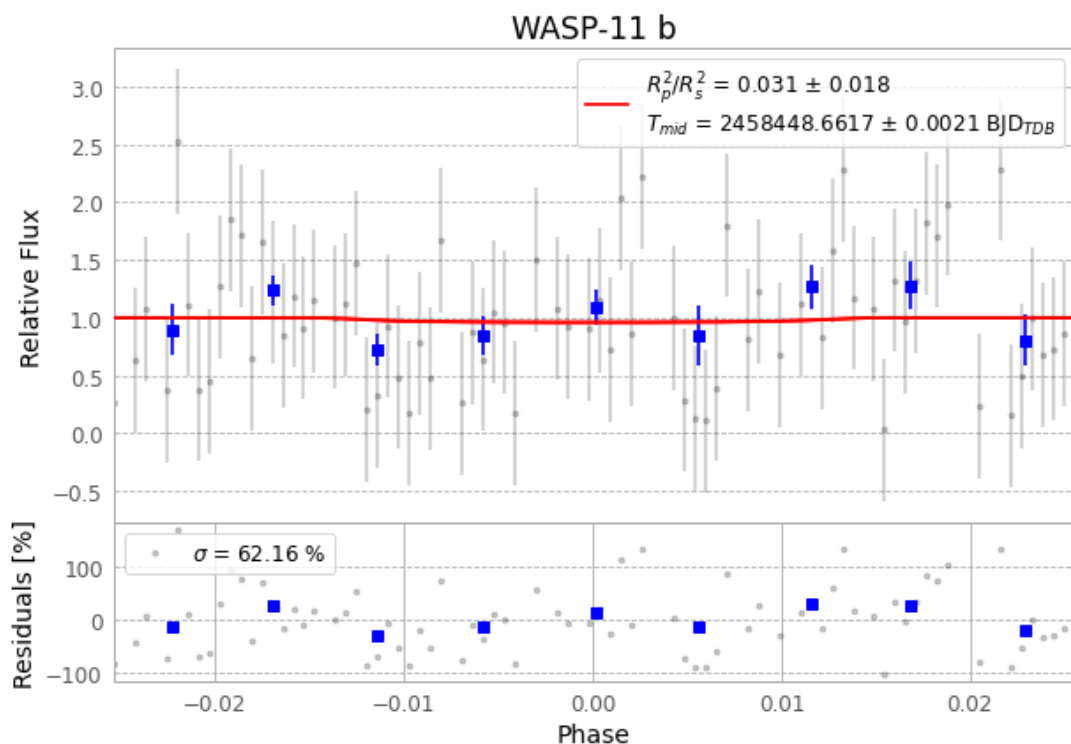


Figure 1 — FinalLightCurve_WASP-11 b_2018-11-25.png (SHA-256 63053b98728071c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [445, 315], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 62578dfdd3d54c36...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

92 FITS frames (60 MB), HATP-10181126012712.FITS → HATP-10181126060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-181126.log (c7ee927d0136...), FinalLightCurve_WASP-11 b_2018-11-25.png (63053b987280...), FinalLightCurve_WASP-11 b_2018-11-25.csv (6c2b0574fb36...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 23:37Z.